Future Directions



Tauhid Zaman

The Personal Agent Race

Jan '26



OpenClaw
Open source

Free

- Open-source agent you run on a Mac mini
- Message it on WhatsApp, Telegram, iMessage

May '26



Gemini Spark
Google

$100/mo

- Google's 24/7 agent
- Lives in Gmail, Docs, Calendar, Drive

Sep '26



Muse
Meta

Free – $100/mo

- Meta's agent for 3B+ users
- Lives in WhatsApp

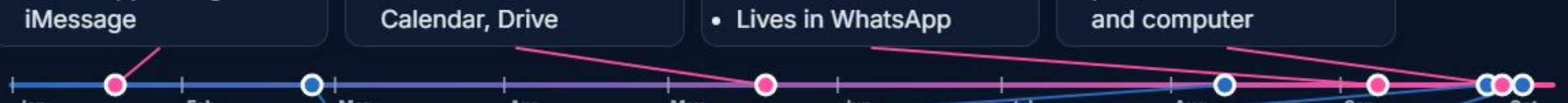
Sep '26



Manus Cue
Manus

Free (early access)

- Each agent gets a wallet, phone number, email, and computer



Feb '26



Hermes Agent
Nous Research

Free

- Open source with long-term memory
- Turns solved tasks into reusable skills

Aug '26



Grok Bot
xAI

$30/mo

- Teams of agents, each with its own computer
- Signs into your apps as you

Sep '26



Instinct
Startup

Invite only

- Text it or call it
- Raised $1B at a $10B valuation

Sep '26



Dots
OpenAI

$100/mo

- OpenAI's always-on agents on GPT-6 Astra
- 4,000+ apps, Slack and Teams

Dots sit at the real launch dates. Price = cheapest monthly plan that includes the agent (Oct 2026). Open-source agents are free but you pay for the AI model.

Who Checks The Agents?

- Agents now send emails, sign into accounts for you, and **spend money**. Mistakes are costly
- Agent certification: prove an agent is safe
- Agent insurance: if the certified agent messes up, the policy pays

Company	What's unique	Founded
AIUC	Certifies agents with ~5,000 attack tests (AIUC-1), re-tests every quarter, then insures them. $40M Series A, Sept 2026	2024
Mount	Liability insurance for deployed agents, with nonstop red-teaming to price the risk	2026
Braintrust	Testing and monitoring tools companies use to check their own agents. $800M valuation	2023
Patronus AI	Runs agents through simulated tasks to catch mistakes before customers do	2023



The Agent Startup Gold Rush

- Pick a job, build an agent that does it, raise at a huge valuation

Startup	What the agent does	Valuation	Numbers
Harvey	Legal work: contracts, due diligence, drafting	**$15.6B**	$550M round, Sept 2026; ARR $400M+ (src)
Sierra	Customer service for brands	**$15B**	$950M round, May 2026 (src)
Instinct	Personal assistant you text or call	**$10B**	$1B round, Sept 2026
Abridge	Writes doctors' clinical notes	**$5.3B**	$300M round, a16z
Decagon	Customer support agents	**$4.5B**	Tender offer, Mar 2026 (src)
Actively	Sales: one agent per customer account	**$250M**	$45M Series B, Apr 2026 (src)



AI In Medicine

- Design drugs on a computer: predict protein shapes, invent new molecules and antibodies
- Predict which drugs will work before running the lab experiment

Company	What's unique	Founded
Isomorphic Labs	DeepMind spinout designing drugs with AlphaFold. Raised $2.1B	2021
Insilico Medicine	AI-found lung drug, now in Phase III trials	2014
Chai Discovery	Designs antibodies from scratch. $3.8B; Lilly, Pfizer, Novartis deals	2024
Xaira	$1B+ launch; protein design from Nobel winner David Baker's lab	2023
Recursion	Robot labs + AI on millions of cell images. 5 drugs in trials	2013



Open Weight Models

- **Open weights**: download the model and run it on your own computer. No per-token bill, and your data stays home
- **The gap is months**: the best open models trail the best closed ones by ~4 months (Epoch AI)
- **China leads**: DeepSeek, Qwen, Kimi, and GLM are the top open models
- **US**: OpenAI released gpt-oss and Google has Gemma. Meta used to release open Llama models, but its newest flagship is closed



What It Takes To Run Them

- Memory ≈ parameters × bytes each: 2 bytes at 16-bit (full precision), 1 at 8-bit, ½ at 4-bit
- (active) = parameters used per word. Fast, but you still load all of them
- One NVIDIA B200 GPU has 192 GB of memory, so an 865 GB model needs 5+ of them
- Click a model to download it from Hugging Face

Model	Maker	Parameters (active)	Memory, 4-bit	Runs on
gpt-oss-20b	OpenAI	21B (3.6B)	**~13 GB**	Laptop with 16 GB
Qwen3.8-27B	Alibaba	27B	**~15 GB**	Laptop or gaming GPU
Gemma 4 31B	Google	31B	**~17 GB**	Laptop or gaming GPU
gpt-oss-120b	OpenAI	117B (5.1B)	**~65 GB**	1 B200
DeepSeek V4-Flash	DeepSeek	284B (13B)	**~150 GB**	1 B200
Mistral Large 3	Mistral	675B (41B)	**~340 GB**	2–3 B200s
DeepSeek V4-Pro	DeepSeek	1.6T (49B)	**~865 GB**	5–8 B200s (one 8-GPU server)
Kimi K3	Moonshot	2.8T (104B)	**~1.56 TB**	16 B200s (8 aren't quite enough)

AI Courses At Yale SOM

Search title, professor, or description | All semesters | All types | All AI levels | 34 classes

MGMT 7206 Financial Econometrics and Machine Learning Part 1 Bryan Kelly
AI focus · fall-2 · PhD
0.5 units · F 1:00 PM–4:00 PM

MGMT 7207 Financial Econometrics and Machine Learning Part 2 Bryan Kelly
AI focus · spring-1 · PhD
0.5 units · Meeting times not listed

MGMT 7405 Algorithms For Democratic Decision-Making Jamie Tucker-Foltz
Uses AI · spring · PhD
1.0 units · Meeting times not listed

MGT 409 AI Foundations for Managers Tauhid Zaman
AI focus · fall-1 · Core
2.0 units · M W 10:05 AM–11:25 AM | M W 1:00 PM–2:20 PM | T Th 10:05 AM–11:25 AM | T Th 1:00 PM–2:20 PM | T Th 2:35 PM–3:55 PM

MGT 413 State & Society Mushfiq Mobarak
Uses AI · fall-2 · Core
2.0 units · T Th 8:30 AM–9:50 AM | T Th 10:05 AM–11:25 AM | T Th 1:00 PM–2:20 PM | T Th 2:35 PM–3:55 PM | T Th 4:10 PM–5:30 PM
Will include discussion on AI issues.

MGT 416 IE: China Zeqiong Huang
Uses AI · spring-1 · Elective
4.0 units · Meeting times not listed

Build Virtual Worlds With AI

- The newest models build 3D worlds, games, and animations from a prompt
- **GPT-6 Astra**
  - Floor plan → Blender model → a walkable Unreal Engine scene (Matt Shumer)
  - "Void Explorer": 2,048 star systems you can fly to and walk on, in the browser (Aituts)
- **Claude Opus 5.5**
  - Full 3D web games from one prompt in one Claude Code session (guide)
  - Prompt gallery: Mario Kart-style racer, endless open world (Tripo)



The Tech To Build Them

Three.js · free

3D in the browser with JavaScript. Yale SOM Virtual is built with it. Share a link and anyone can play.

Blender · free

Model, texture, and animate 3D assets. With Blender MCP, your vibe coder builds the scene for you.

Unreal Engine · free to start

Film-quality games and walkable worlds (Fortnite runs on it). Royalties only after $1M in revenue.

Godot · free, open source

Lightweight game engine for 2D and 3D. We use it in MGT 575 to have agents build games.

Conclusion

- This semester you learned:
  - The basic concepts underlying AI
  - How AI impacts our environment and society
  - How to write code with AI
  - How to build agents with AI
- Final vibe coding session: let's make Yale SOM with AI
- Vibe coding session →

